

Experiment 8: Depth First Search Aim:

Traverse a graph using DFS.

Algorithm:

1. Start.
2. Apply the required AI technique.
3. Generate the solution.
4. Display the result.
5. Stop.

Python Code: graph =

```
{  
'A':['B','C'],  
'B':['D','E'],  
'C':['F'],  
'D':[],  
'E':['F'],  
'F':[]  
}  
visited = set()  
def dfs(node):  
    if node not in visited:  
        print(node,end=" ")  
        visited.add(node)  
        for i in graph[node]:  
            dfs(i)  
dfs('A')
```

Output

```
DFS Traversal:  
A B D E F C
```

Result:

Traversal displayed.